

Unit II

Measure of Central Tendency and Dispersion

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Measures of Central Tendency

In the previous unit we learned about data collection and Tabulation as well as classification of data. We may have to investigate this classified data. Quite possibly the most powerful asset to examine is to calculate a single value which will be the representative of the whole data. This value is called as average or central tendency. Some important definitions of central tendency or average defined by various authors as follows

- i)An average value is a single value within the range of the data that is used to represent all of the values in the series. Since an average is somewhere within range of the data, it is also called a measures of central tendency ----- Croxton & Cowden
- ii)The average is sometime described as a number which is typical of the whole group.----- Leabo
- iii)Average is an attempt to find one single figure to describe whole of figures.-----Clark

From the above definitions we say that a single value which will be representative of whole data and around which a large proportion of observations are clustered are called average or measures of central tendency.

Objectives of Central Tendency

There are two main objectives of central tendency or average

- i) To get a single value for the entire data.
- ii)To facilitate comparison.

Properties of Central Tendency:

- i)It should be simple to understand and easy to calculate.
- ii)It should be rigidly defined.

- iii) This is based on all the observations in the data.
- iv) It should be applicable for quantitative as well as qualitative data.
- v) It should be used for further mathematical calculations.

Types of Central tendency

The following are the important types of central tendency of average

- a) Arithmetic mean
- b) Median
- c) Mode

2.1 Arithmetic Mean(A.M)

The ratio of sum of observations to the total number of observations is called as arithmetic mean or simply mean i.e A.M =

2.1.1. Computations of Arithmetic Mean

a) For Individual Observations or Ungrouped data :

In this case $x_1, x_2, \dots, x_n$ is a set of n observations then arithmetic mean is given by

$$A.M = \frac{x_1 + x_2 + \dots + x_n}{n}$$

A.M is denoted by $\bar{x}$

Example :1

$$\therefore A.M = \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Calculate Arithmetic mean for the following data :13,29,30,16,25,27 .

Answer :

$$A.M = \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$= \frac{13 + 29 + 30 + 16 + 25 + 27}{7}$$

$$= \frac{140}{7}$$

$$\therefore A.M = 23.33$$

Example :2

Calculate Arithmetic mean for the following data :20,26,32,36,40,44,48.

Answer :

$$A.M = \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$= \frac{20 + 26 + 32 + 36 + 40 + 44 + 48}{7}$$

$$= \frac{246}{7}$$

$$\therefore A.M = 35.142$$

b) For Discrete Frequency Distribution :

In this case the observations $x_1, x_2, \dots, x_n$ occur with frequencies $f_1, f_2, \dots, f_n$ the the arithmetic mean is given by ;

$$A.M = \frac{x_1 f_1 + x_2 f_2 + \dots + x_n f_n}{f_1 + f_2 + \dots + f_n}$$

$$= \frac{\sum x_i f_i}{\sum f_i}$$

Where $N = \sum f_i = \text{Total frequency}$

Example :1

Find Arithmetic mean for the following frequency distribution

No.of observation	0	2	3	4	5
Frequency	4	20	25	7	6

Answer :

No.of observation (xi)	Frequency (fi)	xifi
0	4	0
1	20	20
2	25	50
3	7	21
4	6	24
Total	N=62	115

$$A.M = \frac{x_1 f_1 + x_2 f_2 + \dots + x_n f_n}{f_1 + f_2 + \dots + f_n}$$

$$= \frac{\sum x_i f_i}{\sum f_i}$$

$$= \frac{115}{62}$$

$$= 1.8548$$

C)For Continuous Frequency Distribution :

In this case the class intervals along with their frequencies then arithmetic mean is given by ;

$$A.M = \frac{\sum_{i=1}^N f_i x_i}{\sum_{i=1}^N f_i}$$

$$\sum_{i=1}^N f_i x_i$$

Where N = Total frequency and xi's are the mid values of the class interval

Example 1 : Calculate Arithmetic mean for the following frequency distribution

Class Interval	10-30	30-50	50-70	70-90	90-110
Frequency	10	40	80	35	15

Answer:

Class Interval	Frequency (fi)	Mid value (xi)	fixi
10-30	10	20	200

30-50	40	40	1600
50-70	80	60	4800
70-90	35	80	2800
90-110	15	100	1500
Total	N= 180	-----	10900

$$A.M = \frac{\sum_{i=1}^N f_i x_i}{\sum_{i=1}^N f_i}$$

$$\sum_{i=1}^N f_i x_i$$

$$= \frac{10900}{180} = 60.5555$$

Example 2 : Calculate Arithmetic mean for the following frequency distribution.

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	4	7	12	9	8

2.1.2.Properties of Arithmetic Mean

Given that $x = 20$

i) Each value is doubled and then increased by 2 .i.e $y = 2x+2$ then $x = 2x + 2 = (2 \times 20) + 2 = 42$. ii) Each value is increased by 5 and then doubled i.e. $y = 2(x+5)$ then $x = 2(x + 5) = 2(20+5) = 50$

iii) Each value is decreased by 5 i.e. $y = x - 5$ then $x = x - 5 = 20 - 5 = 15$.

Example 2 : Suppose the arithmetic mean of 40 observations is 100. Find arithmetic mean if each observation is i) Increased by 10

ii) Decreased by 5

iii) Doubled

iv) Doubled and then increased by 5

v) Increased by 5 and then doubled

Answer:

In this case we use the property of change of origin and scale

Let x = Original observation and y = New observation

Given that $X = 100$

i) Increased by 10 i.e. $y = x + 10$ then $x = x + 10 = 100 + 10 = 110$

ii) Decreased by 5 i.e. $y = x - 5$ then $x = x - 5 = 100 - 5 = 95$

iii) Doubled i.e. $y = 2x$ then $x = 2x = 2 \times 100 = 200$

iv) Doubled and then increased by 5 i.e. $y = 2x + 5$ then $x = 2x + 5 = (2 \times 100) + 5 =$

205 v) Increased by 5 and then doubled i.e. $y = 2(x+5)$ then $x = 2(x + 5) = 2(100+5) = 210$

Example 3 : Find the combined mean of the following data

Group I	1 = 2100	n1 = 100
---------	----------	----------

Group II	2 = 1500	n2 = 200
----------	----------	----------

Answer : $\bar{x} = \frac{\sum fx}{\sum f}$

$$= \frac{(100 \times 2100) + (200 \times 1500)}{100 + 200}$$

$$= \frac{51000}{300}$$

$$= 1700$$

Example : The mean weekly salary paid to 300 employees of a firm is Rs. 1470. There are 200 men

employees and the remaining are womens .If mean salary of men is Rs.1505.Calculate mean salary of women ?

Answer :

Suppose n_1 = no.of mens and n_2 = no.of womens

Given that $n_1+n_2 = 300$

$$\sum x = 1470$$

$$n_1=200 \therefore n_2=100$$

$$\sum x_1 = 1505$$

$$\sum x_2 = ?$$

$$\begin{aligned} \sum x_c &= \sum x_1 + \sum x_2 \\ \sum x_1 + \sum x_2 \\ 1407 &= \frac{(200 \times 1505) + (100 \times \sum x_2)}{300} \\ 441000 &= 30100 + 100 \sum x_2 \end{aligned}$$

$$\sum x_2 = 1400$$

Example : The arithmetic mean of following frequency distribution having 100 observations is 44.2. Find the missing frequencies.

C.I.	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	5	12	-----	20	----	10	4

Answer :

Given that $N=100$ and $\bar{x} = 44.2$

C.I.	Freq.	Mid value x_i	fix_i
10-20	5	15	75
20-30	12	25	300
30-40	a	35	35a
40-50	20	45	900
50-60	b	55	55b
60-70	10	65	650
70-80	4	75	300

$$\text{Total } N = 51 + a + b \quad 2225 + 35a + 55b \quad N = 100$$

$$51 + a + b = 100$$

$$a + b = 100 - 51$$

$$\therefore a + b = 49 \text{ -----(1)}$$

$$\sum \frac{fix_i}{N}$$

$$\frac{2225}{100} = 22.25$$

$$44.2 = \frac{2225 + 35a + 55b}{100}$$

$$\therefore 35a + 55b = 2195 \text{ -----(2)}$$

Solving equation (1) and (2) simultaneously then we get

a = 25 and b = 24

$\therefore$ Missing frequencies are 25 and 24 respectively.

2.1.3. Merits and Demerits of Arithmetic Mean

Merits:

- i) It is simple to understand and easy to calculate.
- ii) A. M can be rigidly defined.
- iii) It is based on all observations.
- iv) It is capable of further mathematical calculations.

Demerits:

- i) It is applicable only to quantitative data.
- ii) It cannot be determined graphically.
- iii) It cannot be calculated for frequency distribution with an open end class.

2.2. Median

The median is the value separating the higher half from the lower half of sample data. Median divides data into two equal parts. It is also called the positional average.

Definition : Median is the middle most observation in the data set when the observations are arranged in increasing or decreasing order.

2.2.1. Computation of Median

a) For Individual Observations or Ungrouped data :

In this case first we arrange the observation in ascending order.

If the number of observations n is odd then

$$\text{Median} = \frac{n+1}{2} \text{th observation}$$

if the number of observations n is even then

$$\text{Median} = \frac{\left(\frac{n}{2} \right) \text{th observation} + \left(\frac{n}{2} + 1 \right) \text{th observation}}{2}$$

Example 1 : Calculate median for the following data :

18, 49, 37, 24, 28, 11, 44. **Answer :** First we arrange the data in increasing order

11, 18, 24, 28, 37, 44, 49

Here number of observations = n = 7 i.e. odd number

$$\therefore \text{Median} = \left(\frac{n+1}{2} \right) \text{th observation}$$

$$= \left(\frac{7+1}{2} \right) \text{th observation}$$

$$= 4^{\text{th}} \text{ observation}$$

∴ Median= 28

Example 2: Calculate median for the following data : 22,41,21,58,28,32

Answer : First we arrange the data in increasing order

21,22,28,32,41,58

Here number of observations = n = 6 i.e.even number

$$\begin{aligned} \therefore \text{Median} &= \frac{\frac{n}{2}^{\text{th}} \text{ term} + (\frac{n}{2} + 1)^{\text{th}} \text{ term}}{2} \\ &= \frac{3^{\text{rd}} \text{ term} + 4^{\text{th}} \text{ term}}{2} \\ &= \frac{28 + 32}{2} \\ &= 30 \end{aligned}$$

∴ Median =30

Example 3: Calculate median for the following data : 13,11,17,19,15.

Example 4: Calculate median for the following data : 12,8,26,24,18,20

b) For Continuous Frequency distribution :

Procedure

- 1) Calculate less than cumulative frequency (l.c.f.)
- 2) Calculate N/2
- 3) Locate the Median class which is the class corresponding to which cumulative frequency is greater than or equal to N/2 for the first time.
- 4) Apply the following formula of Median

$$\text{Median} = l + \left[\frac{\frac{N}{2} - c.f.}{f} \right] \times h$$

Where l = lower limit of median class

N = Total frequency

c.f.= previous cumulative frequency of median class

f = frequency of median class

h = width of median class

Example:1

Find Median for the following frequency distribution

C.I.	0-5	5-10	10-15	15-20	20-25
Frequency	7	18	25	30	20

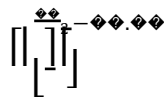
Answer:

C.I.	Frequency	L.C.F.
0-5	7	7
5-10	18	25

10-15	25	$50 = N/2$
15-20	30	80
20-25	20	100
Total	$N = 100$	

Let $N = 100$, $l = 10$, $h = 5$, $c.f = 25$

$\therefore$ Median class is 10-15



Median $= l + x \cdot h$

$$\begin{aligned}
 &= 10 + x \cdot 5 \\
 &= 10 + x \cdot 5 \\
 &= 10 + 1 \cdot 5 \\
 &= 15
 \end{aligned}$$

Example:2 The following incomplete table gives the number of students in different age groups. If the median of the distribution is 11 years, find the missing frequencies.

Age group (year)	0-5	5-10	10-15	15-20	20-25	25-30	Total
No. of students	15	125	-----	66	----	4	300

Answer:

Given that : $N = 300$, Median = 11

Age group	No. of students	l.c.f
-----------	-----------------	-------

0-5	15	15
5-10	125	140
10-15	a	140+a
15-20	66	206+a
20-25	b	206+a+b
25-30	4	210+a+b
Total	N = 210+a+b	

here,

$$N = 300$$

$$210+a+b = 300$$

$$a+b = 90 \text{ ----(1)}$$

$$\text{Median} = l + \left[\frac{\frac{N}{2} - F}{f} \right] \times h$$

∴ Median class is 10-15

$$\text{Median} = l + \left[\frac{\frac{N}{2} - F}{f} \right] \times h$$

$$11 = 10 + \left[\frac{150 - 140}{a} \right] \times 5$$

$$11 = 10 + \left[\frac{10}{a} \right] \times 5$$

$$11 = 10 + \frac{50}{a}$$

$$11 = 10 + \frac{50}{a}$$

$$11a = 10a + 50$$

$$11a - 10a = 50$$

$$a = 50$$

$$a = 50$$

Put a=50 in equation (1)

$$a+b = 90$$

$$50+b = 90$$

$$\therefore b = 90-50$$

$$b = 40$$

$\therefore$ The missing frequencies are 50 and 40 respectively.

2.2.2. Merits and Demerits of Median

Merits:

- i) It is simple to understand and easy to calculate.
- ii) It can be determined graphically.
- iii) It is not affected due to extreme observations.
- iv) It can be computed for a distribution with open end classes.
- v) It is applicable to qualitative data also.

Demerits:

- i) It is not based on all the observations.
- ii) It is not rigidly defined.
- iii) It is not capable of further mathematical treatment.

2.3.Mode

Definition : Mode is the value of variables that occur more number of times in the

data. **2.3.1.Computation of Mode**

a) For Individual Observations or Ungrouped data :

In this case Mode is the value that occurs the maximum number of times of the given data.

Example: Find Mode for the following data:

8, 28, 35, 20, 8 , 25 ,20 ,8

Answer:

Here Mode is the value that occurs the maximum number of times of the given data.

$$\therefore \text{Mode} = 8$$

b) For Discrete Frequency distribution :

In this case the Mode is the value with maximum frequency.

Example: Calculate Mode for the following frequency distribution

X	5	7	9	11	13
f	9	8	15	20	10

Answer :

Here Mode is the value with maximum frequency.

$$\text{Maximum frequency} = 20$$

$$\therefore \text{Mode} = 11$$

c) For Continuous Frequency distribution

In this case first we locate Modal class. The value lies in the class with maximum frequency called

the modal class. Then apply the following formula of mode;

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

L = Lower class limit of the modal class.

f_1 = Frequency of the modal class.

f_0 = Frequency of the class preceding the modal class.

f_2 = Frequency of the class succeeding the modal class.

h = Class interval of the modal class.

Example:

Compute Mode for the following frequency distribution

C.I.	0-10	10-20	20-30	30-40	40-50	50-60
------	------	-------	-------	-------	-------	-------

Freq. 5 7 10 15 12 6

Answer:

C.I.	0-10	10-20	20-30	30-40	40-50	50-60
Freq.	5	7	10=f ₁	15 =f _m	12=f ₂	6

In this example maximum frequency is 15

∴ Modal class is 30-40

$$\begin{aligned} \text{Mode} &= l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h \\ &= 30 + \left[\frac{15 - 10}{2 \times 15 - 10 - 12} \right] \times 10 \\ &= 36.25 \end{aligned}$$

Example:

The daily expenditure of 100 families is given below

Expenditure	20-29	30-39	40-49	50-59	60-69
-------------	-------	-------	-------	-------	-------

No. of families	14	----	27	----	15
-----------------	----	------	----	------	----

If the mode of the distribution is 43.5 , finding the missing frequencies.

Answer:

Given that N = 100 and Mode = 43.5

Expenditure [C.I.]	Class boundaries	No.of families [fi]
20-29	19.5-29.5	14
30-39	29.5-39.5	a
40-49	39.5-49.5	27
50-59	49.5-59.5	b
60-69	59.5-69.5	15
Total		N = 56+a+b

$$N = 100$$

$$\therefore 56+a+b = 100$$

$$\therefore a+b = 100-56$$

$$\therefore a+b = 44 \text{ -----(1)}$$

Here Mode =43.5 therefore modal class is 39.5 -49.5

$$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

$$43.5 = 39.5 + \left[\frac{27 - 14}{2 \times 27 - 14 - 15} \right] \times 10$$

Here we put a+b = 44

$$\therefore a = 44-b$$

$$\therefore 43.5 = 39.5 + \left[\frac{27 - (44-b)}{2 \times 27 - (44-b) - 15} \right] \times 10$$

$$0.4 = \frac{27 - 44 + b}{54 - 44 + b - 15}$$

$$0.4 = \frac{-17 + b}{10}$$

$$4 = -17 + b$$

$$\therefore b = 21$$

Put b= 21 in equation (1)

$$\therefore a+b = 44$$

$$a + 21 = 44$$

$$\therefore a = 44-21$$

$$\therefore a = 23$$

∴ The missing frequencies are 23 and 21 respectively.

Example:

The following table shows the distribution of families according to their daily expenditure. The median and mode for distribution are Rs. 27 and Rs.26 respectively. Find missing frequencies of data

Expenditure(Rs)	0-10	10-20	20-30	30-40	40-50
No. of families	3	----	20	12	-----

Answer:

Given that ;

Median = 27

Mode = 26

Expenditure	No. of families	l.c.f.
0-10	3	3
10-20	a	3+a
20-30	20	23+a
30-40	12	35+a
40-50	b	35+a+b

Given that median= 27 therefore median class is 20-30

$$\text{Median} = l + \left[\frac{\frac{n}{2} - \sum_{j=1}^{k-1} f_j}{f_k} \right] \times h$$

$$\frac{35+a+b}{2} - (3+a)$$

$$27 = 20 + \left[\frac{20}{20} \right] \times 10$$

20

$$\therefore 28 = 35+a+b-6-2a$$

$$\therefore a - b = 1 \text{ -----(1)}$$

Then next given value is Mode i.e. 26 therefore modal class is 20-30

$$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

$$26 = 20 + \left[\frac{20-12}{2 \times 20 - 12 - 12} \right] \times 10$$

$$\therefore a = 8$$

Put a = 8 in equation (1)

$$a - b = 1$$

$$8 - b = 1$$

$$\therefore b = 7$$

$\therefore$ The missing frequencies are 8 and 7 respectively.

Example:

Calculate median and mode for the following data

No. of days absent	Less than 5	Less than 10	Less than 15	Less than 20	Less than 25	Less than 30	Less than 35	Less than 40	Less than 45
No. of students	29	224	465	582	634	644	650	653	655

Answer:

C.I.	l.c.f	Freq.
0-5	29	29
5-10	224	195 f1
10-15	$465 > 327.5$	241 fm
15-20	582	117 f2
20-25	634	52
25-30	644	10
30-35	650	6
35-40	653	3
40-45	655	2
Total		655

$$i) \text{ Median} = l + \left[\frac{\frac{N}{2} - \sum f_1}{h} \right] \times h$$

Let $\frac{N}{2} = 327.5$ therefore median class is 10-15 $\frac{655}{2}$

$$\text{Median} = l + \left[\frac{\frac{N}{2} - \sum f_1}{h} \right] \times h$$

$$= 10 + \left[\frac{327.5 - 224}{241} \right] \times 5$$

$$= 12.1473$$

ii) Mode = $l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$

Here maximum frequency is 241 therefore modal class is 10-15

$$\text{Mode} = l + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

$$= 10 + \left[\frac{241 - 195}{2 \times 241 - 195 - 117} \right] \times 5$$

$$= 11.3529$$

2.3.2. Merits and Demerits of Mode

Merits:

1. It is simple to understand and easy to calculate.
2. It can be determined graphically.
3. It can be calculated for distribution with open end classes.
4. It is applicable for quantitative as well as qualitative data.
5. It is not affected by extreme observations.

Demerits:

1. It is not based on all the observations.
2. It is not rigidly defined.
3. It is not capable of further mathematical treatment.

2.3.3. Relationship among Mean, Median and Mode

A distribution in which mean, median and mode coincide means mean=median=mode is called a symmetric distribution. For moderately asymmetric frequency distribution a very important relationship exists among mean, median and mode called empirical relationships.

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$$

$$\therefore \text{Mode} = 3 \text{ Median} - 2\text{mean}$$

$$\text{And median} = \text{Mode} + (\text{Mean} - \text{Mode})^2_3$$

If we know any of the two values out of three then we compute the third value by using this empirical relationship.

Example:

For a moderately skewed distribution, the arithmetic mean is 10 and mode is 8 then find the median. **Answer:**

Given that

Mean = 10

Mode = 8

Median = ?

In this case we use empirical relationship

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$$

$$\therefore 10 - 8 = 3(10 - \text{Median})$$

$$2 = 30 - 3 \times \text{Median}$$

$$3 \times \text{Median} = 30 - 2$$

$$3 \times \text{Median} = 28$$

$$\therefore \text{Median} = 9.3333$$

Quartiles

Quartiles divide data into four equal parts. There are three quartiles Q_1 , Q_2 , and Q_3 .

Q_1 Q_2 Q_3

$$Q_1 = \text{Lower quartile} = l + \left[\frac{1}{4} \times h \right]$$

$$Q_2 = \text{Second quartile} = \text{Median} = l + \left[\frac{2}{4} \times h \right]$$

$$Q_3 = \text{Upper quartile} = l + \left[\frac{3}{4} \times h \right]$$

Example: Compute quartiles for the following data

34, 11, 16, 23, 6, 9, 25

Answer:

First we arrange data in ascending order

6, 9, 11, 16, 23, 25, 34

Here number of observation = $n = 7$

$$\left(\frac{n}{4} \right)^{th} \text{ observation}$$

i	F
)	i
L	r
o	s
w	t
e	q
r	u
q	a
u	r
a	t
r	i
t	l
i	e
l	=
e	Q_1
=	Q_2

$$= \frac{1}{(n+1)} \left(\sum_{i=1}^n x_i \right)$$

$$= \frac{1}{(7+1)} \left(\sum_{i=1}^7 x_i \right) = \frac{1}{8} (2 + 2 + 2 + 2 + 2 + 2 + 2) = 9$$

ii) Second Quartile = Median = $\frac{1}{2} (x_{(n/2)} + x_{(n/2+1)})$

$$= \frac{1}{2} (2 + 2) = 2$$

$$= \frac{1}{4} \left(\sum_{i=1}^n x_i \right) = \frac{1}{4} (4 + 4 + 4 + 4 + 4 + 4 + 4 + 4) = 16$$

iii) Upper Quartile = Third Quartile = $\frac{3}{4} (x_{(n/2)} + x_{(n/2+1)})$

$$= \frac{3}{4} (2 + 2) = 3$$

$$= \frac{1}{4} \left(\sum_{i=1}^n x_i \right) = \frac{1}{4} (6 + 6 + 6 + 6 + 6 + 6 + 6 + 6) = 25$$

2.4.Measures of Dispersion

We have found in the last chapter of measures of central tendency that central tendency gathers information into a single value. For extra assessment we measure the degree of variation. We realize that average or central tendency gathers information into a single value. But average alone isn't adequate to portray the distribution completely. There might be two distributions with the same mean, however they might not be identical.

Example:

Series A	Series B	Series C
20	20	0
20	25	30
20	25	10
20	10	20
20	20	40
Total = 100	100	100
A.M. = 20	20	20

In the above example arithmetic mean is the same in all three series but they differ in variation. So for further study it is necessary to measure the extent of variation. The important aspect of comparison is dispersion

Some important definitions of dispersion are as follows

1. "The measurement of the scatteredness of the mass of figures in a series about an average is called a measure of variation or dispersion". ----- Simpson & Kafka
2. Dispersion is the measure of the variation of the items. ----- A.L. Bowley
3. Dispersion or spread is the degree of the scatter or variation of the variable about a central tendency. ----- Brooks & Dick
4. It is clear from the above definitions it is clear that the observations are scattered or dispersed from the central value. This variation is called dispersion.

Properties of a good measures of Dispersion

1. It should be simple to understand.
2. It should be easy to calculate.
3. It should be rigidly defined
4. It should be based on each and every observation.
5. It should not be unduly affected by extreme observations.

2.4.1. Range

Definition : Range is the simple method of studying dispersion. Range is defined as the difference between largest and smallest observation.i.e.

$$\text{Range} = L - S$$

Where L = Largest observation

S = Smallest observation

Example:

Compute Range for the following data

40, 22, 69, 58, 68, 28, 95

Answer:

Here Largest observation = 95

And Smallest observation = 22

$$\therefore \text{Range} = L - S$$

$$= 95 - 22$$

$$= 73$$

Example:

Calculate Range for the following frequency distribution

Marks	10-20	20-30	30-40	40-50	50-60
No.of students	6	10	15	5	9

Answer:

Here Largest Observation = 60

And Smallest observation = 10

$$\begin{aligned}\therefore \text{Range} &= L - S \\ &= 60 - 10 \\ &= 50\end{aligned}$$

2.4.1 Merits and Demerits of Range:

Merits of Range:

- i) It is simple to understand and easy to calculate.
- ii) It takes minimum time to calculate the value of Range.

Demerits of Range:

- i) Range is not based on all the observations.
- ii) It does not give a proper idea regarding variation between the extreme observations.

2.4.2. Quartile Deviation

Quartile deviation:

The difference between upper quartile (Q_3) and lower quartile (Q_1) is called inter quartile range and half the inter quartile range is called Quartile deviation.

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

Example:

Find Quartile deviation for the following frequency distribution

C.I.	10-15	15-20	20-25	25-30	30-35
Freq.	2	4	12	8	16

Answer: First we find upper quartile (Q_3) and lower quartile (Q_1)

C.I.	Frequency	L.C.F.
10-15	2	2
15-20	4	6
20-25	12	18
25-30	8	26
30-35	16	42
Total	N=42	

$$i) Q_1 = \text{Lower quartile} = l + \left[\frac{N/4 - c}{h} \right] \times h$$

$$\text{Let } N/4 = 42/4 = 10.5 \therefore$$

Quartile class is 20-25

$$\begin{aligned} Q_1 &= l + \left[\frac{N/4 - c}{h} \right] \times h \\ &= 20 + \left[\frac{10.5 - 6}{5} \right] \times 5 \\ &= 21.875 \end{aligned}$$

$$ii) Q_3 = \text{upper quartile} = l + \left[\frac{3N/4 - c}{h} \right] \times h$$

$$\text{Let } 3N/4 = (3 \times 42) / 4 = 31.5$$

$\therefore$ Quartile class is 30-35

$$\begin{aligned} Q_3 &= l + \left[\frac{3N/4 - c}{h} \right] \times h \\ &= 30 + \left[\frac{31.5 - 26}{5} \right] \times 5 \\ &= 31.7187 \end{aligned}$$

$$\begin{aligned} iii) Q.D. &= \frac{Q_3 - Q_1}{2} \\ &= \frac{31.7187 - 21.875}{2} \\ &= 9.8437 \end{aligned}$$

2.4.3.1. Merits and Demerits of Quartile deviation

Merits :

- i) It is easy to calculate and simple to understand.
- ii) As it takes middle 50% terms hence it is measured better than Range.
- iii) It doesn't include a lot of numerical troubles.

Demerits

- i) Q_1 and Q_3 are both positional measures subsequently not prepared to do advanced

logarithmic treatment.

- ii) If the values are irregular then the result is affected badly.
- iii) It is too much affected by fluctuations of samples.

2.4.4. Variance :

Let X be a random variable whose possible values $x_1, x_2, x_3, \dots$, occur with

$$p_1, p_2, p_3, \dots$$

probabilities $p_1, p_2, p_3, \dots$, respectively. The expected value or arithmetic mean of X, denoted

$$E(X) \text{ or } \mu$$

is defined by

$$E(X) = \mu = x_1 p_1 + x_2 p_2 + x_3 p_3 + \dots + x_n p_n$$

In other words, the mean or expectation of a random variable X is the sum of the products of all possible values of X by their respective probabilities.

Definition : Let X be a random variable whose possible values $x_1, x_2, x_3, \dots$, occur with

$$p_1, p_2, p_3, \dots$$

probabilities $p_1, p_2, p_3, \dots$, respectively. The variance of X, denoted by $\text{Var}(X)$ or σ^2 is defined as

$$\sigma^2 = \text{Var}(X) = \sum_{i=1}^n (x_i - \mu)^2 p_i$$

Deviation : .

2.4.5. Standard

This is the most widely used method of dispersion. The standard deviation was introduced by Karl Pearson in 1823.

The non-negative number **S.D.** σ is called the standard deviation of the

random variable X.

We can also use the simplified form of

$$\sigma^2 = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = \sum_{i=1}^n x_i^2 p_i - \left(\sum_{i=1}^n x_i p_i \right)^2$$

$$= E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \sum_{i=1}^n x_i^2 p_i$$

Example 1 : Three coins are tossed simultaneously, X is the number of heads. Find expected value and variance of X.

Solution : $S = \{\text{HHH}, \text{HHT}, \text{HTH}, \text{HTT}, \text{THT}, \text{TTH}, \text{TTH}, \text{TTT}\}$ and

$$X = \{0, 1, 2, 3\}$$

$X = \text{HHH}$	$P = \frac{1}{8}$	HHH	HHH^2
0	$\frac{1}{8}$	0	0
1	$\frac{3}{8}$	3	3
2	$\frac{3}{8}$	6	12
3	$\frac{1}{8}$	3	9
		$\sum_{i=1}^3 \text{HHH}$ $\frac{12}{8}$ $=$ $\text{HH} = 1$	$\sum_{i=1}^3 \text{HHH}^2$ $\frac{12}{8}$ $=$ $\text{HH} = 1$

$$\begin{aligned} \text{then } E(x) &= \sum_{i=1}^3 \text{HHH} \\ &= \frac{12}{8} \\ &= 1.5 \end{aligned}$$

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$$\begin{aligned} &\sum_{i=1}^3 \text{HHH}^2 \\ &= \left(\text{HH} \right) \text{HH} = 1 \left(\text{HH} \right) \end{aligned}$$

$$\text{Var}(X) = - \text{HH} = 1 = -$$

$$\begin{aligned} &\frac{8(1.5)^2}{24} \\ &= 3 - 2.25 \end{aligned}$$

$$= 0.75$$

Example 2: Find the mean and variance of the number randomly selected from 1 to 15.

Solution :

The sample space of the experiment is $S = \{1, 2, 3, \dots, 15\}$

Let X denote the number selected.

Then X is a random variable which can take values 1, 2, 3, ..., 15. Each number selected is equiprobable therefore

$$P(1) = P(2) = P(3) = \dots = P(15) = \frac{1}{15}$$

$$\begin{aligned} \mu = E(X) &= \sum_{i=1}^{15} i \times \frac{1}{15} \\ &= \frac{1}{15} \times (1 + 2 + 3 + \dots + 15) \\ &= \frac{1}{15} \times \frac{15 \times 16}{2} \\ &= 8 \end{aligned}$$

$$\begin{aligned} \mu = E(X) &= \sum_{i=1}^{15} i \times \frac{1}{15} \\ &= 8 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= \sum_{i=1}^{15} i^2 \times \frac{1}{15} - (\mu)^2 \\ &= \frac{1}{15} \times (1^2 + 2^2 + 3^2 + \dots + 15^2) - 8^2 \end{aligned}$$

$$\begin{aligned} &= \frac{1}{15} \times \frac{15 \times 16 \times 31}{6} - 8^2 \\ &= 82.67 - 64 \\ &= 18.67 \end{aligned}$$

Example 3 : Find expected value and variance of X for the following p.m.f.

(X)	-2	-1	0	1	2
P(X)	0.2	0.3	0.1	0.15	0.25